

Artificial Intelligence (CS-203)

Objectives

- First Order Logic (FOL)
 - Resolution and CNF
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First Order Logic

In propositional logic, we can only represent the facts, which are either true or false. Unfortunately propositional logic is not sufficient to represent the complex sentences or natural language statements. It has very limited expressive power. Consider the following sentence, which we cannot represent using propositional logic.

"Some humans are intelligent", or

"Ali likes cricket."

To represent the above statements we required some more powerful logic, such as *first-order logic*.

Basic Elements of First-order logic:

Constant	1, 2, Sunday, Ahmed, Islamabad, cat,..
Variables	x, y, z, a, b,...
Predicates	Brother, Father, >, <=, ..
Function	sqrt, LeftLegOf,
Connectives	$\wedge$, $\vee$, $\neg$, $\Rightarrow$, $\Leftrightarrow$
Equality	$=$
Quantifier	$\forall$, $\exists$

Atomic sentences

These sentences are formed from a predicate symbol followed by a parenthesis with a sequence of terms.

- Hamid and Akbar are brothers: $\Rightarrow$ Brothers(Hamid, Akbar).
- Chinky is a cat: $\Rightarrow$ cat (Chinky)
- Bill jogged on Sunday $\Rightarrow$ jogged(bill, sunday)

Quantifiers

Universal Quantifier, (for all, everyone, everything)

Universal quantifier is a symbol of logical representation, which specifies that the statement within its range is true for everything or every instance of a particular thing.

The main connective for universal quantifier $\forall$ is implication $\rightarrow$.

Existential quantifier, (for some, at least one)

Existential quantifiers are the type of quantifiers, which express that the statement within its scope is true for at least one instance of something.

The main connective for existential quantifier $\exists$ is and $\wedge$.

In universal quantifier, $\forall x \forall y$ is similar to $\forall y \forall x$.

In Existential quantifier, $\exists x \exists y$ is similar to $\exists y \exists x$.

$\exists x \forall y$ is not similar to $\forall y \exists x$.

Examples:

- All birds fly.
 $\forall x \text{ bird}(x) \rightarrow \text{fly}(x)$.
- Some boys play cricket.
 $\exists x \text{ boys}(x) \rightarrow \text{play}(x, \text{cricket})$.
- Not all students like both Mathematics and Science.
 $\neg \forall (x) [\text{student}(x) \rightarrow \text{like}(x, \text{Mathematics}) \wedge \text{like}(x, \text{Science})]$.
- If it doesn't rain on Monday, Tom will go to the mountains.
 $\neg \text{weather}(\text{rain}, \text{monday}) \rightarrow \text{go}(\text{tom}, \text{mountain})$
- Nobody likes taxes.
 $\neg \exists X \text{ likes}(X, \text{taxes})$
- All basketball players are tall
 $\forall X (\text{basketball_player}(X) \rightarrow \text{tall}(X))$
- There are some x where x is a boy who is intelligent.
 $\exists X \text{ boys}(x) \wedge \text{intelligent}(x)$
- Good boys deserve fruit always
 $\forall X (\text{good_boy}(X) \rightarrow \text{deserve_fruit}(X))$
- All cows eat grass
 $\forall X (\text{cow}(X) \rightarrow \text{eat_grass}(X))$
- Some cows eat grass
 $\exists X (\text{cow}(X) \rightarrow \text{eat_grass}(X))$

- Some cows are not birds and some are

$$\exists x[\text{cow}(X) \wedge \neg \text{bird}(x)] \wedge \exists x[\text{cow}(X) \wedge \text{bird}(x)]$$
- There is no smallest positive real number

$$\forall X [\text{positive}(X) \rightarrow \exists Y[\text{positive}(Y) \wedge \text{less_than}(Y, X)]]$$

Resolution and CNF:

The resolution rule for first-order logic can resolve two clauses if they contain complementary literals, which are assumed to be standardized apart so that they share no variables. Resolution is a single rule of inference that can operate efficiently on a special form of sentences. That special form is called conjunctive normal form (CNF) or clausal form, and has these properties:

- Every sentence is a disjunction (OR) of literals (clauses)
- All sentences are implicitly conjuncted (ANDed).

Steps for Resolution:

1. Conversion of facts into first-order logic.
2. Convert FOL statements into CNF
3. Negate the statement which needs to prove (proof by contradiction)
4. Draw resolution graph (unification).

Example1:

1. The humidity is high or the sky is cloudy.
2. If the sky is cloudy then it will rain.
3. If the humidity is high then it is hot.
4. It is not hot.

and the goal: It-will-rain prove by resolution theorem that the goal is derivable from the knowledge base.

Let us first denote the above clauses by the following symbols.

p = the-humidity-is high, q = the-sky-is-cloudy, r = it-will-rain, s = it-is-hot.

The CNF form of the above clauses thus become

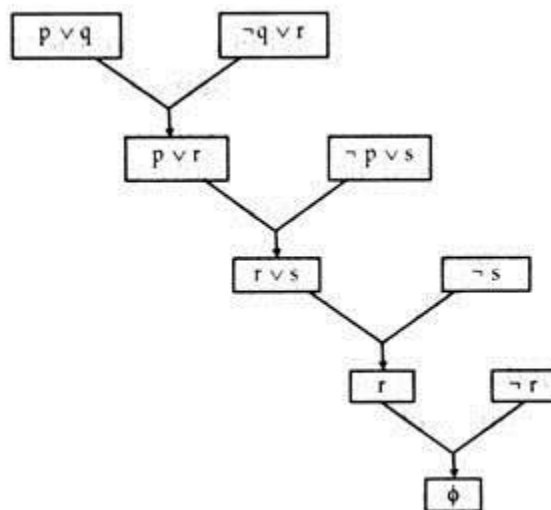
1. $p \vee q$

2. $\neg q \vee r$ (after removing implication)
3. $\neg p \vee s$ (after removing implication)
4. $\neg s$
5. $\neg r$ (negated goal)

When all the clauses are connected through connector $\wedge$ they are called in CNF and conjugated terms for the set S. For example

$$S = p \vee q \wedge \neg q \vee r \wedge \neg p \vee s \wedge \neg s \wedge \neg r$$

Now by resolution algorithm, we construct the graph. Since it terminates with a null clause the goal is proved.



Example2:

Consider the following propositions:

P1: Bilal will sweat today

P2: Today is Humid

P3: Today is hot

P4: Bilal will eat Pakoras

P5: Today is raining

Given with following logical sentences in the knowledge bases.

$P_1 \Leftrightarrow P_2 \vee P_3$
$\neg P_1$
$P_4 \Leftrightarrow \neg P_2 \wedge P_5$
P_4

Using resolution pProve that **today is not humid and today is raining**.

1. $P_1 \rightarrow P_2 \vee P_3$
2. $P_2 \vee P_3 \rightarrow P_1$
3. $\neg P_1$
4. $P_4 \rightarrow \neg P_2 \wedge P_5$
5. $\neg P_2 \wedge P_5 \rightarrow P_4$
6. P_4
7. $\therefore \neg P_2 \wedge P_5$

Using (4) and (6) applying Modus Ponens given inference of (7)

